

Human Capital Theory: Applications to Education and Training

ECON 300 - Labour Economics

Muhebullah Karimzada

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Learning Objectives

- Discuss how the decision to invest in human capital — schooling or on-the-job training — can be analyzed as a standard **investment decision**.
- Explain why the most expensive part of human capital investment is often the **opportunity cost of time**.
- Explain why comparing wages of more- and less-educated workers can be misleading because of:
 - **ability bias**, and
 - the use of education as a **signal** in the labour market.
- Describe why the return to education is large and has increased over time in Canada.
- Discuss when workers or employers should pay for training programs.

Main Questions I

- Why are more-educated workers generally paid more than less-educated workers?
- What factors determine the market rate of return to education?
- If education is such a worthwhile investment, why does not everyone pursue very high levels of education?
- Are more-educated people paid more because education raises productivity, or because education reveals greater inherent productivity?
- What are **signalling** and **screening** in labour markets?

Main Questions II

- How might education serve as a signal?
- If education is mainly a screening device, why might the **private** and **social** returns to education differ?
- Should the government subsidize early childhood learning?
- Are government-funded training programs worth the money?

Big Idea

Human capital theory connects labour economics with investment theory: individuals compare present costs with future benefits.

Human Capital Theory

Definition

Human capital refers to the stock of skills, knowledge, abilities, and training that increases a person's productivity and earning power.

- Investments are made to improve productivity and earnings.
- Costs are incurred today with the expectation of future benefits.
- Rational investment requires that expected benefits exceed expected costs.

Example

A student who spends four years in university gives up current earnings but expects higher wages for decades afterward.

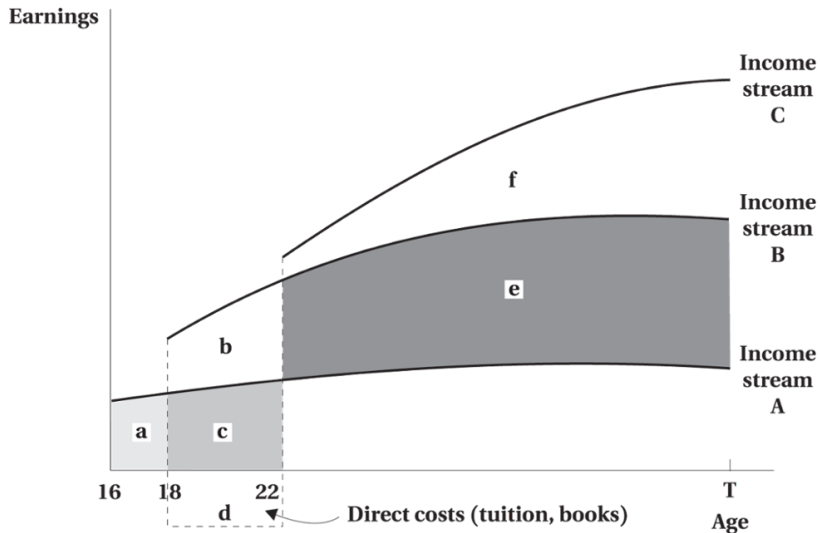
Types of Costs and Benefits

- **Opportunity cost:** forgone income while studying or training.
- **Private costs and benefits:** those borne or enjoyed directly by the individual.
- **Social costs and benefits:** those affecting society as a whole.
- **Real costs:** actual use of resources such as teachers, classrooms, and books.
- **Transfer costs:** payments that reallocate resources, such as taxes or subsidies.

Key Point

The largest cost of schooling is often not tuition, but the earnings the student gives up by not working.

Private Investment in Education



Human Capital Investment: Income Streams

- The vertical axis shows **annual earnings**.
- The horizontal axis shows **age**.
- The figure compares three possible **income streams**:
 - **Income stream A**: entering the labour market early (around age 16).
 - **Income stream B**: completing high school and starting work at age 18.
 - **Income stream C**: attending college/university and entering the labour market around age 22.
- Education changes the entire **lifetime earnings profile**.

Costs of Investing in Education

- Investing in education involves several types of costs.
- **Opportunity costs** (forgone earnings):
 - Area **a**: income forgone between ages 16 and 18 if the individual stays in school.
 - Area **b** + **c**: income forgone between ages 18 and 22 if the individual attends college.
- **Direct costs of education**:
 - Area **d** represents tuition, books, and other schooling expenses.
- These costs occur **while the student is studying** and reduce income early in life.

Benefits of Investing in Education

- Education can increase future earnings.
- **Area b + e** represents the earnings advantage of finishing high school relative to leaving school early.
- **Area f** represents the additional lifetime earnings from college relative to high school.

Investment Decision

Individuals should invest in education if:

$$\text{Present Value of Benefits} > \text{Present Value of Costs}$$

- Human capital theory treats education as a **long-term investment in productivity and earnings**.

Understanding Alternative Income Streams

- Education usually causes earnings to be **lower early in life** because the student is in school.
- Later, education may produce **higher annual earnings**.
- The investment decision depends on whether the later gains outweigh the earlier losses.

Interpretation

Human capital investment changes the entire **age-earnings profile**, not just current income.

Age-Earnings Profiles

- Earnings usually increase with age and experience.
- The increase occurs at a decreasing rate.
- Age-earnings profiles are typically higher for workers with more education.

Example

A university graduate may earn less than a high school graduate at age 20, but more by age 30 or 35.

Optimal Human Capital Investment

Simplifying Assumptions

- Education provides no direct utility or disutility.
- Hours of work are fixed.
- Income streams associated with education are known.
- Individuals can borrow and lend at the real interest rate.

Purpose

These assumptions make the schooling decision a standard **present value maximization problem**.

Formal Analysis of Benefits

Consider an 18-year-old high school graduate deciding whether to work or go to college. The present value of a constant income stream over working years can be written as:

$$PV = Y + \sum_{t=1}^{T-18} \frac{Y}{(1+r)^t} \approx Y + \frac{Y}{r}$$

where:

- Y = annual income,
- r = market interest rate (discount rate),
- T = retirement age.

Interpretation

The higher the discount rate, the less valuable future earnings become today.

Marginal Costs and Benefits of Schooling

- The decision to take **one more year of schooling** depends on:
 - **Marginal cost** of the extra year
 - **Marginal benefit** of the extra year
- The return to schooling can also be summarized by the **internal rate of return**.

Definition

The **internal rate of return** is the discount rate that makes the present value of benefits equal to the present value of costs.

Marginal Cost of One More Year of Education

The cost of one more year of post-secondary education can be written as:

$$MC = Y + D$$

where:

- Y = forgone income while attending school,
- D = direct cost of one more year of school.

Example

If a student gives up \$35,000 in earnings and pays \$7,000 in tuition and books, the marginal cost is \$42,000.

Marginal Benefit of One More Year of Education

The benefit of one more year of schooling is the increase in annual income:

$$MB = \Delta Y$$

The optimal quantity of education is achieved when:

$$MB = MC$$

or equivalently:

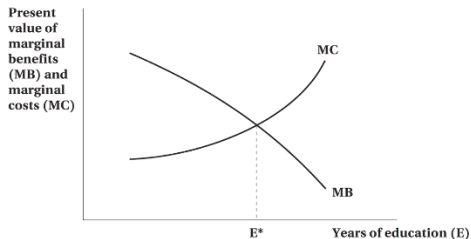
$$i = r$$

where:

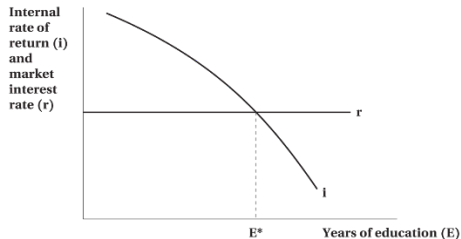
- i = internal rate of return,
- r = market interest rate.

Figure 9.2: Decision Rule for Optimal Human Capital Investment

(a) Marginal benefits equal marginal costs



(b) Internal rate of return equals market interest rate



Decision Rule for Education

To maximize the net present value of lifetime earnings, increase education until:

- the present value of benefits of an additional year equals the present value of costs, or
- $i = r$.

Economic Intuition

- If $i > r$, schooling yields a better return than financial investment.
- If $i < r$, the individual should stop investing in schooling.

Implications of the Theory

- Investment in education should occur early in life because the benefits last longer.
- Individuals expecting interruptions in labour force participation may invest less.
- A progressive tax system can reduce private incentives to invest in education because it reduces after-tax returns.

Example

Someone planning only a short working career has fewer years to recoup the cost of schooling.

Factors Influencing Education

- **Income tax:** greater progressivity lowers the after-tax return to education.
- **Student loans:** reduce borrowing constraints and make schooling more accessible.
- **Government support:** subsidies can increase educational attainment and productivity.

Key Point

Policies affect schooling both by changing the **cost of education** and by changing the **return to education**.

Worked Example: Computing the Return to Education

Mary is an 18-year-old high school graduate who can earn \$30,000 immediately. The market interest rate is 5%, and she plans to work until age 65.

- Without further education, her annual earnings are \$30,000.
- With one more year of schooling, her annual earnings rise to \$33,000.

Question

Should Mary invest in one additional year of education?

Approximating Present Value

A perpetual stream of income Y can be approximated by:

$$PV \approx \frac{Y}{r}$$

This approximation is useful when comparing long-run income streams.

- If $Y = 30,000$ and $r = 0.05$, then

$$PV \approx \frac{30,000}{0.05} = 600,000$$

- If income rises to 33,000,

$$PV \approx \frac{33,000}{0.05} = 660,000$$

Mary's Return to Education

The increase in present value from one more year of schooling is approximately:

$$660,000 - 600,000 = 60,000$$

The textbook example reports that the extra year of schooling raises the present value of future income by about \$23,943 using a finite working-life horizon.

The internal rate of return can be approximated by:

$$i = \frac{\Delta Y}{Y + D}$$

- If $D = 0$:

$$i = \frac{3000}{30,000} = 10\%$$

- If $D = 3000$:

$$i = \frac{3000}{33,000} \approx 9\%$$

Interpreting the Worked Example

- If the internal rate of return exceeds the market interest rate, schooling is a worthwhile investment.
- Mary should invest because:
$$10\% > 5\%$$
- Direct costs would need to become much larger before the investment would stop being worthwhile.

Interpretation

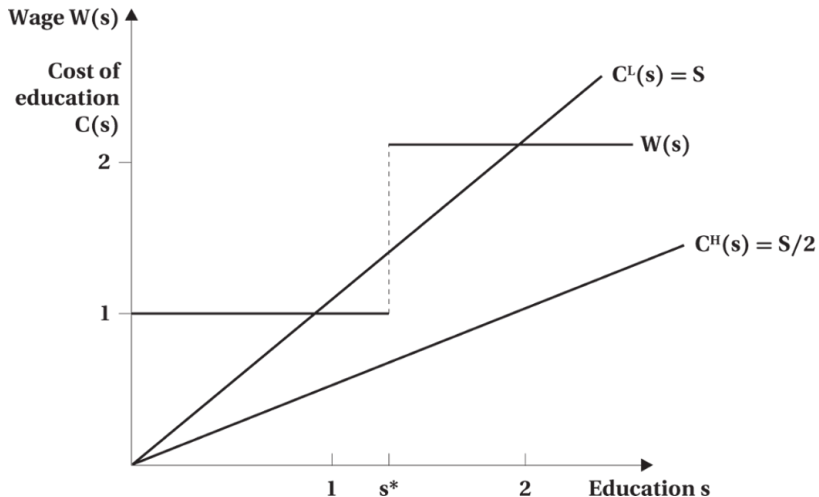
Education is attractive here because the earnings gain is large relative to both forgone income and direct costs.

Definition

Education may act as a **filter**, **signal**, or **screening device** if it helps employers identify worker productivity rather than directly increasing productivity.

- Employers may offer higher wages to educated workers because they believe education signals higher productivity.
- In this view, education helps sort workers rather than necessarily making them more productive.

Figure 9.3: Offered Wage and Signalling Cost Schedules



Interpreting Signalling and Screening

- **Signalling:** workers obtain education to signal desirable traits such as ability, persistence, or discipline.
- **Screening:** firms use education to sort applicants.
- If signalling is very important, the **private return** to education may exceed the **social return**.

Example

An employer may interpret a university degree as evidence that the applicant is more productive, even if the job uses only part of what was learned.

Empirical Evidence: Education and Earnings

- Earnings increase with age and experience.
- The increase is usually most rapid up to middle age.
- The earnings gap between education groups tends to widen over the life cycle.

Interpretation

This pattern is consistent with human capital accumulation and higher long-run rewards to education.

Figure 9.4: Earnings by Age and Education

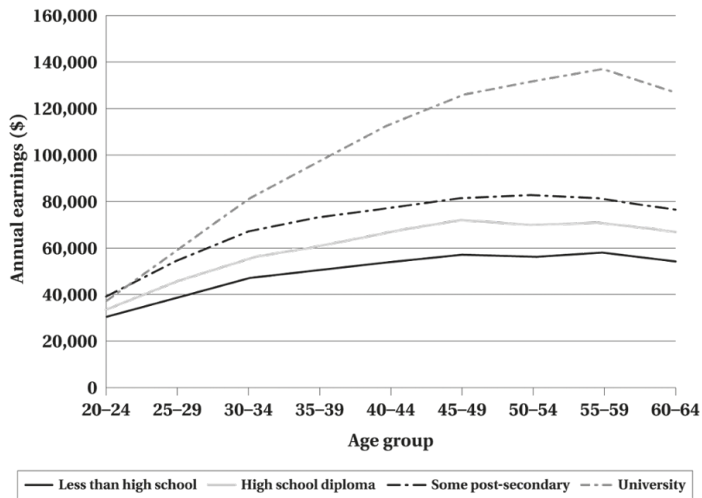


Table 9.1: Estimates of the Private Returns to Schooling in Canada

Table 9.1 Estimates of the Private Returns¹ to Schooling in Canada, 2000		
Level of Schooling	Males	Females
Bachelor's degree ²	12	14
Master's degree	3	5
Ph.D.	nc ²	4
Medicine	21	22
Bachelor's Degree by Field of Study	Males	Females
Education	9	14
Humanities and fine arts	nc	10
Social sciences ³	11	14
Commerce	9	19
Natural sciences	9	8
Engineering and applied science	9	14
Health sciences	18	18

Reading Table 9.1

- Returns differ across:
 - level of schooling,
 - gender,
 - field of study.
- Professional degrees such as medicine often have especially high returns.
- Returns to master's and PhD degrees are often smaller on the margin because those programs involve higher costs and longer time out of the labour market.

Human Capital Earnings Function

A common empirical specification is:

$$\ln Y = \alpha + rS + \beta_1 EXP + \beta_2 EXP^2 + \varepsilon$$

where:

- Y = earnings,
- S = years of schooling,
- EXP = experience,
- r = estimated return to schooling.

Interpretation

The coefficient on schooling, r , approximates the percentage increase in earnings from one additional year of education.

Figure 9.5: Log Earnings by Years of Schooling

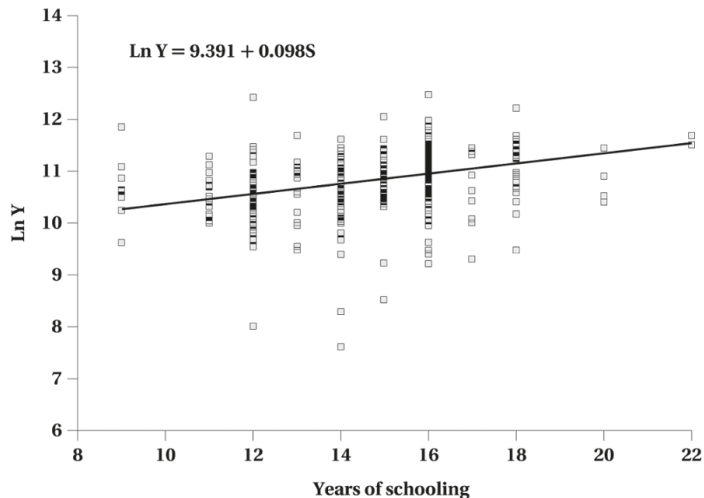


Table 9.2: Estimated Returns to Schooling and Experience

Table 9.2 Estimated Returns to Schooling and Experience, 2015 (dependent variable: log annual earnings)		
	Men	Women
Intercept	9.232 (714.15)	8.605 (628.29)
Years of schooling	0.080 (105.31)	0.114 (140.77)
Experience	0.054 (75.96)	0.041 (62.51)
Experience squared	− 0.0009 (60.14)	− 0.0006 (43.85)
R-squared	0.136	0.204
Sample size	121,947	99,398

Interpreting the Earnings Function Results

- The schooling coefficient is positive for both men and women.
- Experience raises earnings, but at a decreasing rate:
 - positive coefficient on experience,
 - negative coefficient on experience squared.
- This produces the familiar concave age-earnings profile.

- It is difficult to control for:
 - innate ability,
 - motivation,
 - perseverance,
 - tolerance for effort.
- These factors can bias estimated returns to education.
- This is known as **ability bias**.

Economists use several strategies:

- **Natural experiments**
- **Twin studies**
- **Compulsory school attendance laws**
- **Proximity to college** as an instrument for schooling

Goal

These methods try to separate the effect of education from unobserved ability.

What Does the Evidence Suggest?

- The initial belief that returns to education were greatly overstated by ability bias appears to have been exaggerated.
- Education still seems to have a substantial causal effect on earnings.
- The measured returns discussed are usually **private monetary returns**.

Important Distinction

Private returns and social returns need not be the same.

Social Returns to Education

- Social returns capture benefits beyond the individual's higher wages.
- Education may improve:
 - civic participation,
 - health,
 - lower crime,
 - innovation,
 - productivity spillovers.
- Natural experiments based on regional differences in education are often used to estimate social returns.

Increased Returns to Education and Inequality

- Increases in the return to education have occurred alongside broader increases in income inequality.
- However, the link is not complete.
- Inequality also depends on:
 - unemployment risk,
 - within-group wage dispersion,
 - top-end earnings growth.

Increasing Returns to Education Since 1980

- There was little change in returns to schooling in the 1980s and early 1990s.
- Returns increased substantially after the mid-1990s.
- Demand-side factors appear important:
 - skill-biased technological change,
 - globalization,
 - changing occupational structure.
- Wage polarization and very high top incomes also matter.

Definition

Training is an investment in skills acquired after labour market entry.

- **General training:** useful in many firms.
- **Specific training:** useful mainly in the firm providing the training.

- Skills are transferable across firms.
- Other firms are willing to pay more for workers with these skills.
- Therefore workers are usually willing to bear much of the cost.

Example

Learning general software, accounting, or coding skills often raises wages across many employers.

Specific Training

- Skills are valuable mainly to the current employer.
- Other firms will not pay a premium for these skills.
- Therefore the firm often pays at least part of the training cost.

Example

Training on a firm's proprietary software or internal production process is often specific training.

Figure 9.6: Costs, Benefits, and Financing of Training

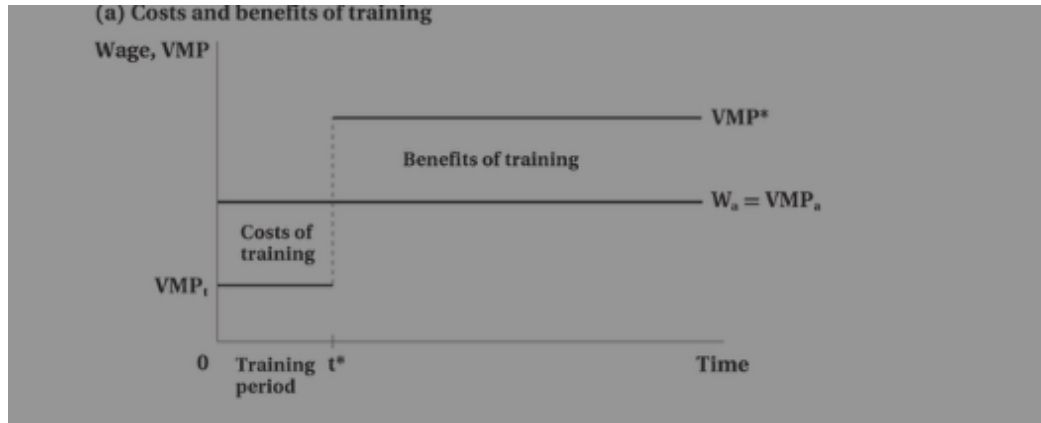


Figure 9.6: Costs, Benefits, and Financing of Training

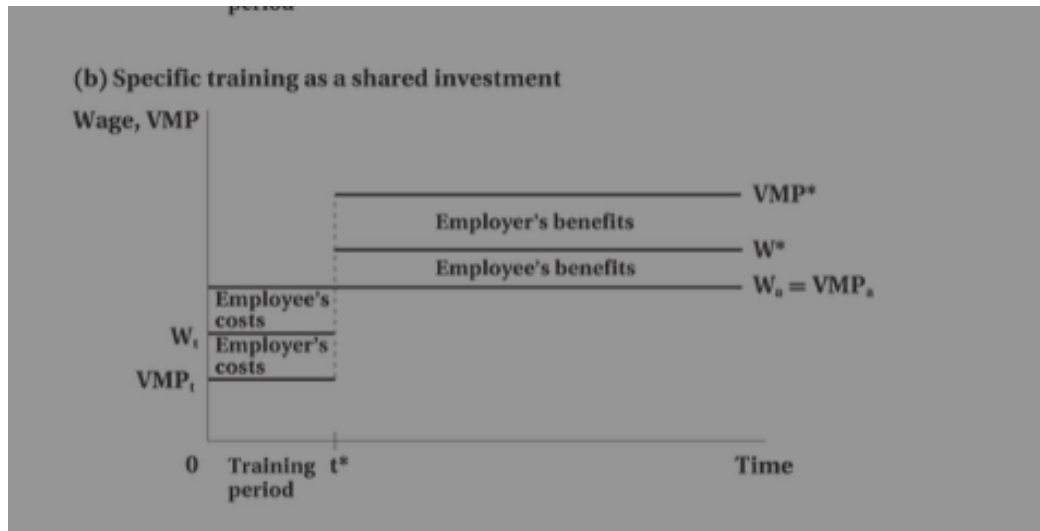
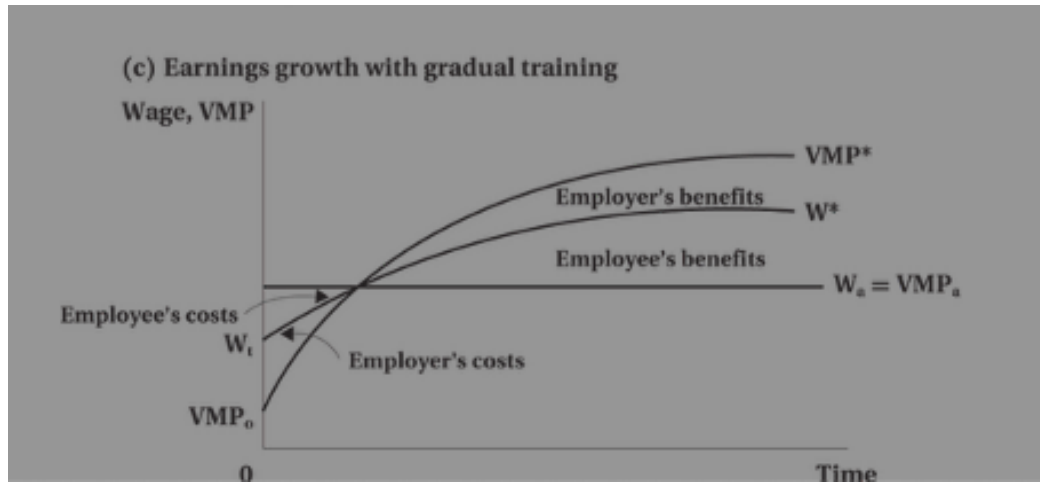


Figure 9.6: Costs, Benefits, and Financing of Training



Who Should Pay for Training?

- **General training:** workers usually pay, often through lower trainee wages.
- **Specific training:** firms often pay, because the productivity gains are tied to that firm.
- In practice, costs may be shared when both parties benefit.

Training and the Role of Government

- Private markets may provide less than the socially optimal amount of training because of:
 - imperfect information,
 - regulatory restrictions,
 - borrowing constraints,
 - labour market disadvantage.
- Training subsidies targeted to disadvantaged workers may:
 - increase working hours,
 - raise wages,
 - reduce poverty.

- Education and training are investments in **human capital**.
- The schooling decision compares present costs with future earnings benefits.
- The opportunity cost of time is often the largest cost of education.
- Human capital theory and signalling theory offer different explanations for the education–earnings link.
- Empirical evidence suggests that returns to education are substantial, though estimation must address ability bias.
- General and specific training differ in who should pay for them.
- Government may play a role when private markets underinvest in human capital.

Thank You!